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MIL-C-18832(NOrd)

24 June 1955

MILITARY SPECIFICATION

CONNECTOR, BREAKAWAY ELECTRO-MECHANICAL

1 SCOPE

1.1 SCOPE - This specification describes the requirements for design, procurement, manufacture, inspection and test of a breakaway electro-mechanical connector for linking a torpedo to an electrical cable.

2. APPLICABLE DOCUMENTS

The following specifications, standards, drawings and publications, either in their entirety or in part, as is indicated herein, of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

QQ-B-611	Brass, commercial: bars, plates, rods, shapes, sheets and strips.
QQ-B-626	Brass, leaded and non-leaded (copper-zinc alloy) rods, bars, shapes and forging.
QQ-P-416	Plating, cadmium (electrodeposited).
QQ-S-633	Steel, carbon, bars (general purpose)
QQ-S-763	Steel, corrosion resisting, bars and forgings (except for reforging).
QQ-S-766	Steel, corrosion resisting, plates, sheets, strips and structural shapes.

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MILITARY

JAN-P-658	Packaging and Packing of Electrical Equipment and Spare Parts.
MIL-P-14	Plastic Materials, Molding and Plastic Part, Molded, Thermosetting.
MIL-T-5422	Testing; Environmental, Aircraft Electronic Equipment.

NAVY DEPARTMENT

NAVORD OS 6341 Miscellaneous General Ordnance Design Requirements.

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 Marking for Shipment.

DRAWINGS

Bureau of Ordnance List of Drawings No. 477149 and matl. listed thereon.

Bureau of Ordnance List of Drawings No. 477150 and matl. listed thereon.

Bureau of Ordnance Drawing No. 1146395

Bureau of Ordnance Drawing No. 1146396

PUBLICATIONS

NAVY DEPARTMENT

OP 400 General Instructions for the Design, Manufacture and Inspection of Naval Ordnance Equipment.

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

2.2 OTHER PUBLICATIONS - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 Screw-Thread Standards for Federal Services. (Applications for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 GENERAL - A breakaway electro-mechanical connector shall consist of three sub-assemblies:

(a) A receptacle
(b) A plug
(c) A lanyard clip

For the purposes of this specification, these shall be defined as follows:

3.1.1 RECEPTACLE - The receptacle is defined as the sub-assembly which is mounted on a torpedo. It shall contain electrical contacts which permit energizing torpedo circuits from an external source and a plunger which can be actuated by a lanyard to mechanically control internal circuits of a torpedo.

3.1.2 PLUG - The plug is defined as the sub-assembly which is normally attached to a free-swinging electrical cable. It shall be equipped with electrical contacts which mate with those of the receptacle. It shall be designed to permit coupling with the receptacle over a previously installed lanyard clip.

3.1.3 LANYARD CLIP - The lanyard clip is defined as the sub-assembly, the removal of which from the receptacle actuates the receptacle plunger. It shall be so designed that it may be clipped onto the receptacle independently of the plug.

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3.1.4 UNCOUPLING FORCE

3.1.4.1 PLUG AND RECEPTACLE - Uncoupling of the plug and receptacle shall be effected by a tension force of 35 ± 15 lbs. applied within 60° of the axis of the connector.

3.1.4.2 LANYARD CLIP AND RECEPTACLE - Uncoupling of the lanyard clip from the receptacle shall be effected by a tension force of 35 ± 15 lbs. applied within 60° of the axis of the connector.

3.2 MATERIALS

3.2.1 METALS - All metal parts shall be made from materials meeting the requirements of specifications or drawings listed in Section 2.

3.2.2 MOLDED PLASTIC INSERTS - All molded plastic inserts shall be made of materials which conform to Specification MIL-P-14, Type MTE Plastic Materials, Molding, and Plastic-Part, Molded, Thermosetting.

3.2.3 FUNGUS RESISTANT MATERIALS - Preference shall be given in the selection of materials to those which are inherently funginert, in accordance with paragraph 3.3.3 of NAVORD OS 6341, Miscellaneous General Ordnance Design Requirements.

3.2.4 SURFACE CHARACTERISTICS - Breakaway electrical connectors shall have surfaces which are not cracked, warped, chipped or blistered. Connector surfaces which normally come in contact with cable or conductor insulation shall be rounded so as to prevent cutting or abrasion of the insulating material.

3.2.5 DIELECTRIC STRENGTH - Inserts shall meet requirements for dielectric strength as given in 4.5.3.

3.3 DESIGN AND CONSTRUCTION

3.3.1 GENERAL

3.3.1.1 DIMENSIONS - Dimensions shall be as shown on the following Bureau of Ordnance Drawings:

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Receptacle - BuOrd L.D. No. 477150
Plug - BuOrd L.D. No. 477149
Lanyard clip- BuOrd L. D. No. 477149

3.3.1.2 POLARIZING - The plug and receptacle shall be so designed as to assure proper contact alignment before any electrical connection between contacts is made.

3.3.2 RECEPTACLE

3.3.2.1 PRESSURE The receptacle, when mounted on the shell of a torpedo, shall not leak under an hydrostatic pressure of 600 psi. The plunger shall not be actuated by this pressure.

3.3.2.2 CONTACTS - Eight contacts shall be positioned and lettered as shown on Bureau of Ordnance Drawing 1146395. Each contact shall be capable of carrying 2 amperes at 115 volts AC (rms) continuously and 8 amperes at 115 volts AC (rms) for a period of 30 seconds without damage.

3.3.2.3 SAFETY-PIN - The receptacle shall be equipped with a removable safety pin to prevent accidental actuation of the plunger. The receptacle shall be designed to prevent removal of the safety pin unless the lanyard clip is in place.

3.3.2.4 DUST COVER - The receptacle shall be provided with a dust cover which can be attached when the plug or lanyard clip is not in place.

3.3.3 PLUG

3.3.3.1 CONTACTS - Eight contacts shall be positioned and lettered as shown on Bureau of Ordnance Drawing 1146396. Each contact shall be capable of carrying 2 amperes at 115 volts AC (rms) continuously and 8 amperes at 115 volts AC (rms) for a period of 30 seconds without damage.

3.4 ENVIRONMENTAL REQUIREMENTS

3.4.1 VIBRATION - When coupled together, the breakaway electro-mechanical connector shall be capable of withstanding vibration with excursion, frequency ranges and duration, as shown below, in each of three mutually perpendicular planes without uncoupling or opening of electrical connections.

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Total Excursion	Frequency Range	Duration
0.060 ± 0.006 inch	10-30 cps	1 hour
0.009 ± 0.001 inch	31-60 cps	1 hour
0.005 ± 0.0005 inch	61-100 cps	1 hour

3.4.2 TEMPERATURE

3.4.2.1 OPERATING TEMPERATURE - The breakaway electro-mechanical connector shall be capable of exhibiting specified performance over an ambient temperature range of -40°F to +130°F.

3.4.2.2 STORAGE TEMPERATURE - The breakaway electro-mechanical connector shall be capable of exhibiting specified performance after storage over an ambient temperature range of -65°F to + 160° F.

3.4.3 HUMIDITY - The breakaway electro-mechanical connector shall be capable of exhibiting specified performance in atmospheres with relative humidities of 95 per cent maximum.

3.4.4 MOISTURE AND CORROSION RESISTANCE - All parts of the breakaway electro-mechanical connector shall be made of a corrosion resistant material or shall be protected with suitable corrosion resistant treatment in accordance with NAVORD OS 6341, Miscellaneous General Ordnance Design Requirements, and the detailed requirements specified in the Bureau of Ordnance Drawings.

3.4.5 ACCELERATION - The breakaway connector shall be capable of exhibiting specified performance after being subjected to acceleration of 15g for a duration of 0.2 seconds and with a waveshape approximating a half sine-wave in each of three mutually-perpendicular planes. Such accelerations shall not cause uncoupling of the connector.

3.4.6 SAND AND DUST - The breakaway connector and the receptacle equipped with the dust cover shall be capable of exhibiting specified performance after exposure to sand and dust conditions as defined in Specification MIL-T-5422, Testing; Environmental, Aircraft Electronic Equipment.

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3.5 DIMENSIONS AND TOLERANCES - Where dimensions are given in this specification without specific tolerances or where no dimensions are shown, all tolerances and dimensions on the manufacturer's drawings or permitted by their shop practices shall be considered acceptable.

3.6 GAGES - The contractor shall provide whatever gages and inspection equipment are necessary and adequate for manufacturing use to insure that the material to which this specification applies will meet the requirements shown on the applicable drawings. If so stipulated in the contract, requisition, or order, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractors inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.7 THREADS - Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28 (latest issue) Screw-Thread Standards for Federal Services. The class or fit for threads shall be as specified on the drawings and the NOT-GO gaging practice shall be as set forth in the Supplement to the aforementioned Handbook H28.

3.8 MARKING - Unless otherwise specified in the contract, requisition or order, the breakaway connector shall be marked with the manufacturer's name or trade-mark and the appropriate Bureau of Ordnance Drawing number. Marking of the lanyard clip shall be optional.

3.9 REQUESTS FOR DEVIATION OF MATERIALS AND PROCESSES Requests for use of materials and processes other than herein specified or deviations in design or configuration of parts from that shown on the applicable drawing shall be submitted to the Bureau of Ordnance via the cognizant inspector prior to use in manufacture. Such requests shall specify or define the proposed material and processes in sufficient detail to permit determination as to their acceptability for the application concerned.

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3.10 WORKMANSHIP

3.10.1 GENERAL - All parts of the breakaway connector shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be paid to neatness and thoroughness of soldering, marking of parts and assemblies, plating, painting, riveting, machine-screw assemblage, and freedom of parts from burrs and sharp edges.

3.10.2 CLEAN-UP - Units shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign material after final assembly. Burrs and sharp edges as well as rosin flash which might crumble shall be removed.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 SAMPLING AND INSPECTION

4.1.1 SAMPLING - Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government Inspector in the determination of the acceptability of products submitted by a supplier for Government inspection will be in accordance with the provisions of MIL-STD-105.

4.1.2 INSPECTION

4.1.2.1 GOVERNMENT INSPECTION - Government inspection shall be in accordance with the provisions of "General Procedures for Government Inspection" as contained in OP 400.

4.1.2.2 CONTRACTOR'S INSPECTION - Contractor's inspection shall be in accordance with the provisions of "Inspection by Contractor" as contained in OP 400.

4.1.2.3 INSPECTION LOT - As applied to Government inspection of units of product, the term "lot" shall mean "inspection lot", i.e., a collection of units of product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" shall be as determined by the Government Inspector and may differ from the quantity designated in the contract or order as a lot for production, shipment or other purpose.

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4.2 CLASSIFICATION OF TESTS - The inspection and testing of the connector shall be as follows:

(a) Preproduction Tests: Preproduction tests are those tests accomplished on samples which are representative for the production of the item to determine that the design of the production meets the requirements of the specification.

(b) Inspection Tests: Final production tests are those tests accomplished on equipment submitted for acceptance.

4.3 PREPRODUCTION TESTS - Preproduction tests will be performed by an activity designated by the Government or the Government Inspector. Preproduction test samples shall consist of 20 connectors.

4.3.1 TEST PROCEDURE - Sample connectors shall be subjected to all tests listed in Table I, in the order shown. After completion of the group I tests, the sample shall be divided into 4 parts and subjected to the remaining tests as indicated in Table I. The tests shall be carried to completion regardless of failures.

4.3.2 ACCEPTANCE CRITERIA - Acceptance criteria for the preproduction tests shall not exceed one failure for Group I and shall be in accordance with MIL-STD-105, code letter C, AQL 4.0%, for each of Groups II, III, IV, and V. Failure to comply with these test procedures may result in rejection of the lots submitted for test or cessation of production as determined by the procuring activity.

Table I

PREPRODUCTION TESTS

Test	Procedure
Group I (all connectors)	
Visual and Mechanical Inspection	4.5.1
Pressure	4.5.2
Dielectric Strength	4.5.3
Contact Current-carrying capacity	4.5.4
Uncoupling Force	4.5.5 and 4.5.6
Group II	
High Temperature	4.5.7.1
Low Temperature	4.5.7.2
Dielectric Strength	4.5.3

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Group III	
Moisture and Corrosion Resistance	4.5.8
Dielectric Strength	4.5.3
Sand and Dust	4.5.12
Group IV	
Vibration	4.5.10
Fungus Resistance	4.5.11
Dielectric Strength	4.5.3
Group V	
Acceleration	4.5.9
Dielectric Strength	4.5.3
Uncoupling Force	4.5.5 and 4.5.6

4.3.3 DISPOSITION OF MATERIAL - Disposition of material submitted for preproduction tests shall be requested from the contracting agency.

4.4 INSPECTION TESTS

4.4.1 TEST CONDITIONS - Unless otherwise specified, the connector shall be subjected to tests under the following conditions:

Temperature	Room Ambient
Altitude	Normal Ground
Humidity	Room Ambient to 95 percent relative maximum

4.4.2 INSPECTION FACILITIES AND TEST EQUIPMENT - The contractor shall furnish and maintain all necessary facilities and equipment for performing all inspection tests. The test equipment shall be adequate in quantity and unless definite requirements are stated herein, or on the pertinent drawings, of sufficient accuracy and quantity to permit performance of the required tests.

4.4.3 MANUFACTURER'S TESTS - The contractor shall conduct inspection tests as specified in 4.4.4 to assure that each connector assembly is in compliance with the requirements of section 3 before submission to the Government Inspector for acceptance.

4.4.4 TEST PROCEDURE - Connectors shall be subjected to the inspections listed in Table II in the order shown.

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Table II

PRODUCTION TESTS

Test	Procedure
Visual and Mechanical Inspection	4.5.1
Pressure	4.5.2
Dielectric Strength	4.5.3
Uncoupling Force	4.5.5 and 4.5.6

4.5 TEST INSTRUCTIONS

4.5.1 VISUAL AND MECHANICAL INSPECTION - Connectors shall be inspected to verify that the materials, design, construction, physical dimensions, marking and workmanship are in accordance with the requirements of this specification.

4.5.2 PRESSURE TEST - The receptacle, without safety pin or lanyard clip, shall be installed in a suitable test fixture and subjected to a hydrostatic pressure of 600 psi for five minutes. The pressure shall be increased from zero at a rate not to exceed 100 psi per second. Pressure shall be applied in the direction corresponding to that experienced in service. There shall be no indication of the insert's being purshed out of the shell and no change in the condition of the switches. Within one hour after completion of this test, the receptacle shall be capable of withstanding the dielectric strength test, 4.5.3.

4.5.3 DIELECTRIC STRENGTH TEST - An alternating potential of commercial frequency and amplitude of 1000 volts rms shall be applied between mutually insulated terminals and between each terminal and the shell of the connector. The test voltage shall be increased from zero at a rate not to exceed 500 volts per second until 1000 volts is reached and then maintained at that value for at least 30 seconds and not more than one minute. Connectors shall withstand this test without arcing, breakdown or damage.

4.5.4 CURRENT CARRYING CAPACITY - The connector shall be wired and coupled as for actual service, and shall be connected to an AC source to draw 8 amperes at 115 volts $\pm$ 10% through each contact. After 30 seconds, the load shall be removed, the connector uncoupled and the contacts examined. The connector shall

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then be recoupled and connected again to the same current source. The current shall be adjusted to 2 amperes and the loads shall be non-inductive. The connector shall be uncoupled and re-coupled not oftener than once each 10 seconds for a total of 50 operations. At the end of this test, the contacts shall show no undue pitting or burning.

4.5.5 UNCOUPLING FORCE (PLUG) - The receptacle shall be rigidly mounted in a manner which will permit coupling and uncoupling the plug without imparting movement to the receptacle. The plug shall be uncoupled by a tension force applied within 60° of the axis of the connector. Tension shall be measured on the axis and at four equally spaced points 55° ± 5° from the axis, making four measurements at each point for Preproduction Tests (Table I) and one measurement at each point for Production Tests (Table II). The force determined from each of these measurements shall be 35 ± 15 lbs.

4.5.6 UNCOUPLING FORCE (LANYARD CLIP) - The receptacle shall be rigidly mounted in a manner which will permit coupling and uncoupling the clip without imparting movement to the receptacle. The lanyard clip shall be uncoupled by a tension force applied within 60° of the axis of the connector. Tension shall be measured on the axis and at four equally spaced points 55° ± 5° from the axis, making four measurements at each point for Preproduction Tests (Table I) and one measurement at each point for Production Tests (Table II). The force determined from each of these measurements shall be 35 ± 15 lbs.

4.5.7 TEMPERATURE

4.5.7.1 HIGH TEMPERATURE - The assembled connector shall be placed in a test chamber and the temperature of the chamber raised to 165°F. After the temperature of the chamber has stabilized at this value, it shall be held for 48 hours. At the conclusion of this period, the temperature of the chamber shall be lowered to 130°F. When the receptacle has reached this temperature, it shall be capable of meeting the tests specified in 4.5.3, 4.5.5 and 4.5.6, except that in the latter two paragraphs, only one measurement shall be made at each position.

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4.5.7.2 LOW TEMPERATURE - The assembled connector shall be placed in a test chamber and the temperature of the chamber lowered to -65°F. After the temperature of the chamber has stabilized at this value, it shall be held for 48 hours. At the conclusion of this period, the temperature shall be raised to -40°F. When the connector has stabilized at this temperature, it shall be capable of meeting the tests specified in 4.5.3, 4.5.5 and 4.5.6, except that in the latter two paragraphs, only one measurement shall be made at each position.

4.5.8 MOISTURE AND CORROSION RESISTANCE - The connector shall be exposed to environment simulating that at dockside in a position protected from spray for a period of 10 ± 1 days. At the end of this period the plunger shall move freely and the connector shall be able to withstand the dielectric strength test of 4.5.3.

4.5.8.1 SALT SPRAY TEST - As an alternate to 4.5.8, the salt spray test specified in MIL-T-5422, Testing; Environmental, Aircraft Electronic Equipment, may be performed, except that the duration of the test shall be limited to 48 ± 5 hours. At the end of this period, the plunger shall move freely, and the connector shall be capable of withstanding the dielectric strength test of 4.5.3.

4.5.9 ACCELERATION - The receptacle shall be mounted rigidly on the testing machine. The unsupported length of the cable attached to the plug and the lanyard clip shall not exceed 4 feet. The connector shall then be subjected to accelerations of 15g with a duration of 0.2 seconds and a waveshape approximating a half sine wave in each of three mutually perpendicular planes. The connector shall not uncouple. Momentary opening of electrical contacts shall be acceptable.

4.5.10 VIBRATION - The connector shall be mounted rigidly on the testing machine. The unsupported length of cable attached to the plug and the lanyard clip shall not exceed 4 feet. The connector shall be subjected to the vibration schedule of 3.4.1. In each frequency range, the frequency shall be varied between the limits given at a uniform rate over a 3 to 5 minute period for the duration of the test. The connector shall remain coupled and no interruption of electrical contact shall occur.

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4.5.11 FUNGUS RESISTANCE - Test procedure shall be in accordance with Specification MIL-T-5422, Testing; Environmental, Aircraft Electronic Equipment.

4.5.12 SAND AND DUST - The connector shall be subjected to the test specified in Specification MIL-T-5422, Testing; Environmental, Aircraft Electronic Equipment, except that sand and dust velocity through the chamber shall be between 100 and 200 feet per minute.

5. PREPARATION FOR DELIVERY

5.1 DOMESTIC SHIPMENT - Unless otherwise specified in the order or contract, packaging and packing for domestic shipment shall be in accordance with approved commercial packing methods, adequate to protect the connector assembly from injury and to assure safe delivery at destination. The method used shall be commensurate with shipping at lowest practicable commerical rates.

5.2 OVERSEAS SHIPMENT - Unless otherwise specified, for overseas shipment, the connector shall be packaged for shipment in accordance with Specification JAN-P-658, Packaging and Packing of Electrical Equipment and Spare Parts.

5.3 Unless otherwise specified in contract or order, labeling and marking of all shipping containers shall be in accordance with pertinent drawings and MIL-STD-129, Marking for Shipment.

6. NOTES

6.1 INTENDED USE - The breakaway electro-mechanical connector is intended for use between torpedoes and launching vessels (surface and air). It provides electric connections for energizing torpedo circuits from the launching vessel and a mechanical system for energizing the torpedo circuits from its own power source.

6.2 ORDERING DATA - Procurement data should specify title, number and date of this specification.

PATENT NOTICE - When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the

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